At a glance · for Ahmed · Oct 2026

CMS TEAM Impact Report: Full Source Audit

Rainfall Health report by Christina Keny, PhD
Published Jul 27, 2026 (footer reads "May 2026")
104 claims checked against primary sources

Scorecard

74 of 104 **claims need a fix**
wrong + unsupported + internal-only + outdated; ~9 verified claims still need a citation

41 Wrong	**17** Unsupported	**9** Unverifiable (internal)
7 Outdated	**28** Verified	**2** Not checked

How the bad citations got through

- **Jun 25:** the first numbered draft (Word) already had the bad refs: 13 refs incl. Tarazi, Becher, Bandeen-Roche, MedPAR, CMS-5540-F and NSQIP.
- **Jun 26:** Kyle's Source & Citation Review flagged 4: Tarazi (no such paper), AAMC (superseded), Becher (wrong locator), Bandeen-Roche (corrupted).
- **Jun 29:** 2 swapped (Tarazi → Ho 2022, Becher → Etzioni 2003), 1 cut (Bandeen-Roche), 3 added (Deiner, Gill, Tang). **No number was fixed.**
- **To print:** MedPAR, CMS-5540-F and NSQIP went out word for word, just renumbered (8, 9, 11).
- **Jul 1:** the cover rewrite moved "first mandatory", "$1.2M Year 1" and "30% by PY5" from the body onto the cover.

Top 6 must-fix (audit priority order; badge = flagged before publication)

1 Risk cap is 20%, not ±30% Cover, p.2, p.7, Fig 4 No
Track 3 caps gains and losses at 20% every year; max spread is 40 points, not 60.
CMS-1808-F · 42 CFR 512.550

2 Ref 3: Ho 2022 has no 74% p.3, p.11 Yes
The 74% / 83.7% over 85 comes from Hewitt 2016 (UK cohort).
Hewitt 2016

3 AAMC: 2018 numbers, 2024 citation p.3 Yes
Use up to 86,000 physicians by 2036, incl. up to 19,900 surgeons.
AAMC 2024

4 "40% of inpatient surgeries" p.3 + stat box Yes
Etzioni's 40% is all discharges; say ~37% of surgical stays.
HCUP #281

5 Ref 9: CMS-5540-F doesn't exist p.11 No
The TEAM final rule is CMS-1808-F (89 FR 68986).
CMS-1808-F

6 California: 107, not 110 p.2, p.8 Yes
107 hospitals in 11 metro areas; the regions printed sum to 118.
CMS TEAM list

Checks out, keep (with citation): 721 hospitals (CMS list) · $481M CMS savings target · 56% and $1.2M (Advisory Board) · 188 CBSAs and the 30-day episode (CMS-1808-F) · 19.4% surgery prevalence (Bicket 2024) · Rainfall company facts (MedCity). **Not checked:** 1980 "26 million"; "stacked on other P4P programs".

Full list of 22 issues on page 2 · corrected references on page 3

All 22 issues · audit priority order

All 22 issues and how to fix them

● Red = wrong ● Amber = unsupported, internal-only or outdated ● Green = verified

#	Issue (report pages)	Verdict	Pre-pub flag	Fix	Source link(s)
1	Risk cap stated as "±30% / 30% of Medicare revenue by PY5"; "60-point spread" p.1, 2, 7	Wrong	No	Track 3 caps gains and losses at 20% in every year (2026–2030); max spread is 40 points. Fix cover, p.2, p.7, Fig 4.	CMS-1808-F · 42 CFR 512.550
2	Ref 3: 74% / >80% over 85 cited to Ho 2022, which has neither p.3, 11	Wrong	Yes	Cite Hewitt 2016 (BMJ Open) and add "in a UK cohort".	Hewitt 2016 · Ho 2022
3	AAMC "121,000 by 2030 / 20–30k surgeons" are 2018 numbers, cited to the 2024 report p.3, 11	Wrong	Yes	Use the 2024 report: up to 86,000 physicians by 2036 (13,500–86,000), up to 19,900 surgeons.	AAMC 2024 · report PDF
4	"40% of inpatient surgeries": Etzioni's 40% is all hospital discharges (1999) p.3, 11	Wrong	Yes	Say "more than a third (~37%) of inpatient surgical stays"; cite HCUP #281; stat box ~37%.	HCUP #281 · Etzioni 2003
5	Ref 9 "CMS-5540-F" doesn't exist; the TEAM rule is CMS-1808-F p.11	Wrong	No	CMS-1808-F. Fed Regist. 2024;89(167):68986–70046. Aug 28, 2024.	CMS-1808-F · rule PDF
6	California "110 hospitals / 26 counties / LA 59"; regions sum to 118 p.2, 8	Wrong	Yes	107 mandated hospitals in 11 metro areas; LA isn't mandatory (Cedars-Sinai voluntary); rebuild regions.	CMS TEAM list · Rainfall CA
7	Fig 3 "max win/loss per case" are ACS/IAC 2023 averages; 204,014 is their case count p.6, 7	Wrong	Partly	Retitle "Average gain vs loss per case (2023, uncapped)"; "~204,000 qualifying Medicare episodes in 2023 (ACS/IAC)".	ACS/IAC analysis
8	Ref 8 credits a CMS MedPAR LOS study that doesn't exist p.3, 11	Wrong	Yes	Cite AJRR 2024 (3.0 → 1.1 days) and Burnett 2023 (57.2% of Medicare TKA billed outpatient).	AJRR 2024 · Burnett 2023
9	Ref 11 "ACS NSQIP procedure-mix" not found; Fig 2 "majority of all five" contradicts its chart p.5, 11	Wrong	No	Cite HCUP #281, add superscript 11; spinal fusion ~30–40%; source or drop 78% / 47% / 58%.	HCUP #281
10	Ref 12 title doesn't exist; "Year 1", $14M and "Optum – $6M" aren't Optum's p.1, 3, 6, 7	Wrong	Partly	Cite Nation R, et al., Advisory Board (56%, $1.2M); drop "Year 1"; label $14M / –$6M as Rainfall estimates.	Advisory Board
11	One $14.5M, three meanings and attributions (p.6, p.7 "Optum", p.9 peak) p.6, 7, 9	Unverifiable	No	Pick one definition; attribute to Rainfall's internal model (ref 14) with a method note; drop "Optum".	*Internal (ref 14)*
12	"First mandatory surgical episode model" / "first wave"; CJR was mandatory p.1, 2, 3	Wrong	No	"CMS's first mandatory multi-procedure surgical episode model"; drop "first wave".	CMS-1808-F (CJR, 89 FR 69636)
13	CJR-X "Oct 2027, 2,500+ hospitals" p.3	Outdated	No	Finalized (CMS-1849-F): starts Jan 1, 2028, ~2,200 hospitals (2,217 on Sept 30 list).	CMS-1849-F · CJR-X list
14	Fig 1 uses 2017 Census projections; 21.0% / 22.4% match neither series p.3, 4	Outdated / Wrong	No	Redraw with 2023 series (71.2 / 78.3 / 82.1M; 20.6 / 22.0 / 22.8%); "exceed 20% by 2030"; cite Census.	Census 2023, Table 2
15	Refs 4–6 back frailty and dementia only, not polypharmacy or social vulnerability p.3, 11	Unsupported	No	"Frailty, dementia and polypharmacy..."; add Hewitt for polypharmacy; update Gill to final issue.	Gill 2022 · Tang 2020
16	19.4% is right (Bicket 2024), but its ref was deleted, not corrected p.3, 5	Verified, uncited	Yes	Add numbered ref: Bicket MC, et al. Ann Surg Open. 2024;5(2):e421.	Bicket 2024
17	p.10 "Platform results": up to 20%, $85M, 70–80% are Rainfall's own claims or nothing p.2, 10	Unverifiable / Unsupported	No	Label $85M as a Rainfall projection; "±20% of target price"; drop 70–80% and "documented gains".	MedCity, Feb 20, 2026
18	"Proven Results / Case examples" (p.8) shows no results p.2, 8	Unsupported	No	Retitle "California: where TEAM hits hardest"; drop "Proven Results", or add real cases.	*No source to cite*
19	Disclaimer says "publicly available data", but refs 13–14 are internal p.11	Wrong	No	"...public data, published sources, and internal Rainfall Health analyses (refs 13–14)".	*Internal (refs 13–14)*
20	p.7 "up to 20% at risk by PY3": Track 3 is 20% from PY1 p.7	Wrong	No	"...up to 20% at risk from PY1 (Track 3)".	CMS-1808-F
21	p.5 three "highest" claims, garbled and uncited p.5	Unsupported	No	"High post-acute spending (22–63% of episode cost[9])", or drop.	*Uses ref 9 (CMS-1808-F)*
22	Cosmetic: "$14.5" no "M"; "May 2026" footer; CPHQ; mixed ref fonts p.1–11	Wrong (minor)	Partly	"$14.5M"; footer "July 2026"; CPHQ consistent; one font for refs 1–15.	—

Flagged before publication: Yes 6 · Partly 3 · No 13 of 22

Reference list · published numbering

Corrected references

Each corrected citation is a clickable link.
Text fixes for each claim are on page 2.

Ref	As published	Problem	What's wrong	Corrected citation (link)
1	Census, "Vintage Population Estimates & National Population Projections, 2020–2050"	Wrong title	Not a real product name; number (71M) is right but means age 65+, not "eligible for Medicare".	U.S. Census Bureau. 2023 National Population Projections Tables: Main Series, Table 2. Nov 2023.
2	Etzioni DA, et al. Ann Surg. 2003;238(2):170-177	Doesn't back claim	Real paper, but its 40% is all hospital discharges (1999), not surgeries. Swapped in for Becher on Jun 29.	McDermott KW, Liang L. HCUP Statistical Brief #281. AHRQ; 2021. (~37% of surgical stays)
3	Ho VP, et al. JAMA Surg. 2022;157(6):499-506	Doesn't back claim	No 74% and no over-85 figure. Replaced the fake Tarazi ref on Jun 29.	Hewitt J, et al. BMJ Open. 2016;6(3):e010126. (UK cohort)
4–6	Deiner 2014 · Gill 2022 · Tang 2020	Partly backs claim	Added Jun 29 "to be safe"; cover frailty and dementia only. Gill needs its final issue.	Gill TM, et al. JAMA Surg. 2022;157(12):e225155. · Tang 2020 · Deiner 2014 unchanged
7	AAMC. Complexities of Physician Supply and Demand... 2021 to 2036. 2024	Ref OK, text wrong	Citation is real; the text quotes the 2018 report (121,000 by 2030).	AAMC 2024 projections · report PDF: up to 86,000 by 2036, incl. up to 19,900 surgeons
8	CMS. Inpatient LOS and outpatient migration trends... MedPAR	Doesn't exist	No such CMS study. Carried word for word from the Jun 25 draft.	AJRR 2024 Annual Report; Burnett RA, et al. J Arthroplasty. 2023;38(6):992-997.
9	CMS. TEAM Final Rule (CMS-5540-F). Aug 28, 2024	Wrong rule number	CMS-5540 was a 2023 RFI. Carried from Christina's May 5 Sources line.	CMS-1808-F. Fed Regist. 2024;89(167):68986-70046. · PDF
10	CMS. FY 2027 IPPS Proposed Rule, CJR-X. Apr 10, 2026	Outdated	Superseded by the final rule (Aug 4, 2026): start Jan 1, 2028.	CMS-1849-F (final). Fed Regist. 2026;91(148):49570-50461.
11	ACS NSQIP & Medicare claims procedure-mix data, 2023–2024	Not found	No such dataset found; no in-text superscript in the published report.	HCUP Statistical Brief #281 (as ref 2); add the in-text 11 at Fig 2
12	Advisory Board / Optum. "TEAM Financial Impact Model". 2025	Title doesn't exist	56% and $1.2M are real; "Year 1", $14M, ± $14.5M, –$6M are not Optum's.	Nation R, et al. What hospitals need to know about TEAM for 2026. Advisory Board, Nov 25, 2025. · ACS/IAC TEAM analysis (2023 data)
13	Rainfall. California TEAM-Mandated Hospitals Brief. Apr 2026	Internal, not found	Not found in Drive or Slack; Rainfall's own public page says 107, not 110.	CMS TEAM participant list (Aug 18, 2026 file) · Rainfall CA page (107)
14	Rainfall. California Episode Economics, 17-System model. Apr 2026	Internal, not found	Model file not found; also cited for product capabilities on p.10.	*Get the model file from Christina, or label it "internal Rainfall analysis".*
15	MedCity News. Feb 20, 2026 (Yahoo Feb 18)	Verified	Company facts check out.	Plescia M. MedCity News, Feb 20, 2026 (add URL and author)
New	(19.4% stat on p.3 and p.5 is uncited)	Add ref	Bandeen-Roche was cut on Jun 29 instead of corrected.	Bicket MC, et al. Prevalence of Surgery Among Individuals in the United States. Ann Surg Open. 2024;5(2):e421.

Refs 13–14 are internal Rainfall analyses (California brief, 17-system model): not public, so they can't be checked. The disclaimer must name them (issue #19).